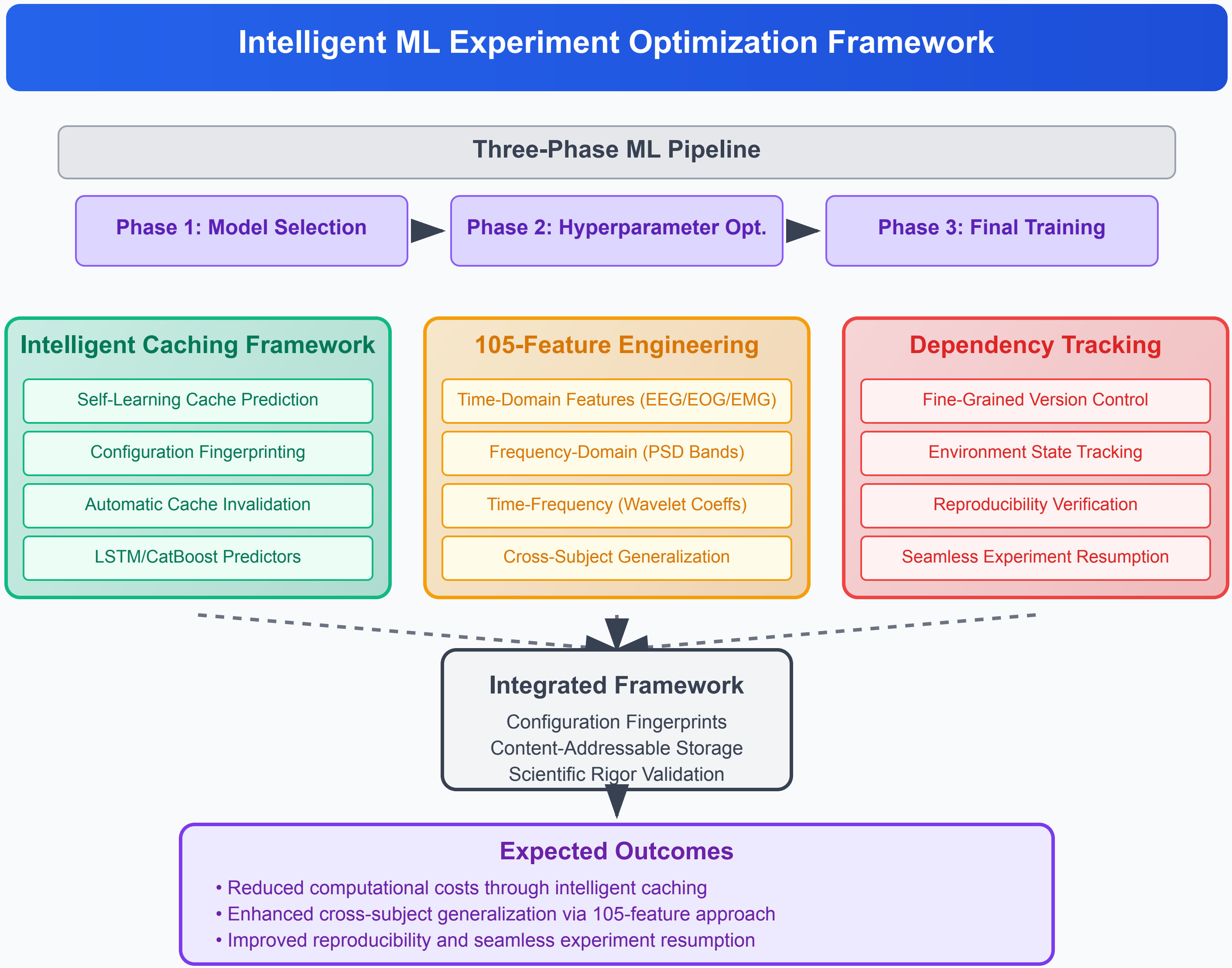
Intelligent ML Experiment Optimization Framework
Three-Phase ML Pipeline
Phase 1: Model Selection
Phase 2: Hyperparameter Opt.
Phase 3: Final Training
Intelligent Caching Framework
Self-Learning Cache Prediction
Configuration Fingerprinting
Automatic Cache Invalidation
LSTM/CatBoost Predictors
105-Feature Engineering
Time-Domain Features (EEG/EOG/EMG)
Frequency-Domain (PSD Bands)
Time-Frequency (Wavelet Coeffs)
Cross-Subject Generalization
Dependency Tracking
Fine-Grained Version Control
Environment State Tracking
Reproducibility Verification
Seamless Experiment Resumption
Integrated Framework
Configuration Fingerprints
Content-Addressable Storage
Scientific Rigor Validation
Expected Outcomes
• Reduced computational costs through intelligent caching
• Enhanced cross-subject generalization via 105-feature approach
• Improved reproducibility and seamless experiment resumption